Introduction to Machine Learning

AI Learning Hub - Comprehensive Course Guide

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What is Machine Learning?

Machine Learning (ML) is a subset of artificial intelligence that gives systems the ability to learn and improve from experience without being explicitly programmed. ML focuses on developing algorithms that can access data, learn from it, and make predictions or decisions. There are three main types of ML: supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning uses labeled datasets to train algorithms to classify data or predict outcomes. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines (SVM).

Unsupervised learning works with unlabeled data, discovering hidden patterns. Key algorithms include K-Means Clustering, DBSCAN, Principal Component Analysis (PCA), and Hierarchical Clustering.

Feature Engineering and Data Preprocessing

Feature engineering is the process of using domain knowledge to create features that make machine learning algorithms work better. This includes:

1. Handling Missing Values: Imputation strategies like mean, median, mode, or KNN-based imputation.
2. Feature Scaling: StandardScaler (z-score normalization) and MinMaxScaler.
3. Encoding Categorical Variables: One-hot encoding, label encoding, target encoding.
4. Feature Selection: Filter methods (correlation), wrapper methods (RFE), embedded methods (Lasso regularization).
5. Dimensionality Reduction: PCA, t-SNE, UMAP for reducing feature space.

Model Evaluation and Metrics

Evaluating ML models requires proper metrics:

Classification Metrics: Accuracy, Precision, Recall, F1-Score, AUC-ROC curve, Confusion Matrix. Precision measures the proportion of true positives among predicted positives. Recall measures the proportion of true positives among actual positives.

Regression Metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R-squared (R2) score.

Cross-validation: K-fold cross-validation splits data into k subsets, training on k-1 and validating on the remaining fold. This reduces overfitting and gives a more reliable estimate of model performance.